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*Tensor contraction of  $a$  and  $b$  along specified axes and outer product.*

## Function Signatures

```
tf.tensordot( a, b, axes, name=None )
```

## Code Examples

```
tf.tensordot(  
a, b, axes, name=None  
)
```

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a, b, axes, name=None  
)
```

## Documentation

Tensor contraction of  $a$  and  $b$  along specified axes and outer product.

## View aliases

Main aliases

`tf.linalg.tensordot`

See

Migration guide for  
more details.

`tf.compat.v1.linalg.tensordot`, `tf.compat.v1.tensordot`

`Tensordot` (also known as tensor contraction) sums the product of elements from `a` and `b` over the indices specified by `axes`.

This operation corresponds to `numpy.tensordot(a, b, axes)`.

Example 1: When `a` and `b` are matrices (order 2), the case `axes=1` is equivalent to matrix multiplication.

Example 2: When `a` and `b` are matrices (order 2), the case `axes = [[1], [0]]` is equivalent to matrix multiplication.

Example 3: When `a` and `b` are matrices (order 2), the case `axes=0` gives the outer product, a tensor of order 4.

Example 4: Suppose that  $(a_{ijk})$  and  $(b_{lmn})$  represent two tensors of order 3. Then, `contract(a, b, [[0], [2]])` is the order 4 tensor

$c_{\{jklm\}}$  whose entry

corresponding to the indices  $(j,k,l,m)$  is given by:

$$c_{\{jklm\}} = \sum_i a_{\{ijk\}} b_{\{lmi\}}.$$

In general,  $\text{order}(c) = \text{order}(a) + \text{order}(b) - 2 \cdot \text{len}(\text{axes}[0])$ .

## Returns

## Raises